



Case Study Session 5: Valuing Environmental, Social, and Governance (ESG): Financing Constraints and Workplace Safety

3,130/4,130 Corporate Finance

Key messages of this case study

1. You will learn the basic idea of **ESG**, i.e., environmental, social and governance factors, in corporate finance.
2. You will discuss an empirical research paper related to ESG.
 - This paper looks at the relationship between financing constraints and workplace safety.
 - You will learn how to cope with endogeneity in corporate finance and how to focus on causality rather than correlation.
 - You will get an introduction on how to structure and write an empirical paper in financial economics which is particularly important for students intending to write a bachelor thesis in finance.

ESG

1. Recap: What is ESG?
2. How is ESG measured?
3. Why is ESG important in corporate finance?

Recap: What is ESG?

ESG aims to incorporate environmental, social, and governance (ESG) factors in the investment decision of investors so that such investments facilitate firms' effort to run their businesses sustainably.

Environmental

- climate change – including physical risk and transition risk
- resource depletion, including water
- waste and pollution
- deforestation

Social

- working conditions, including slavery and child labor
- local communities, including indigenous communities
- conflicts
- health and safety
- employee relations and diversity

Governance

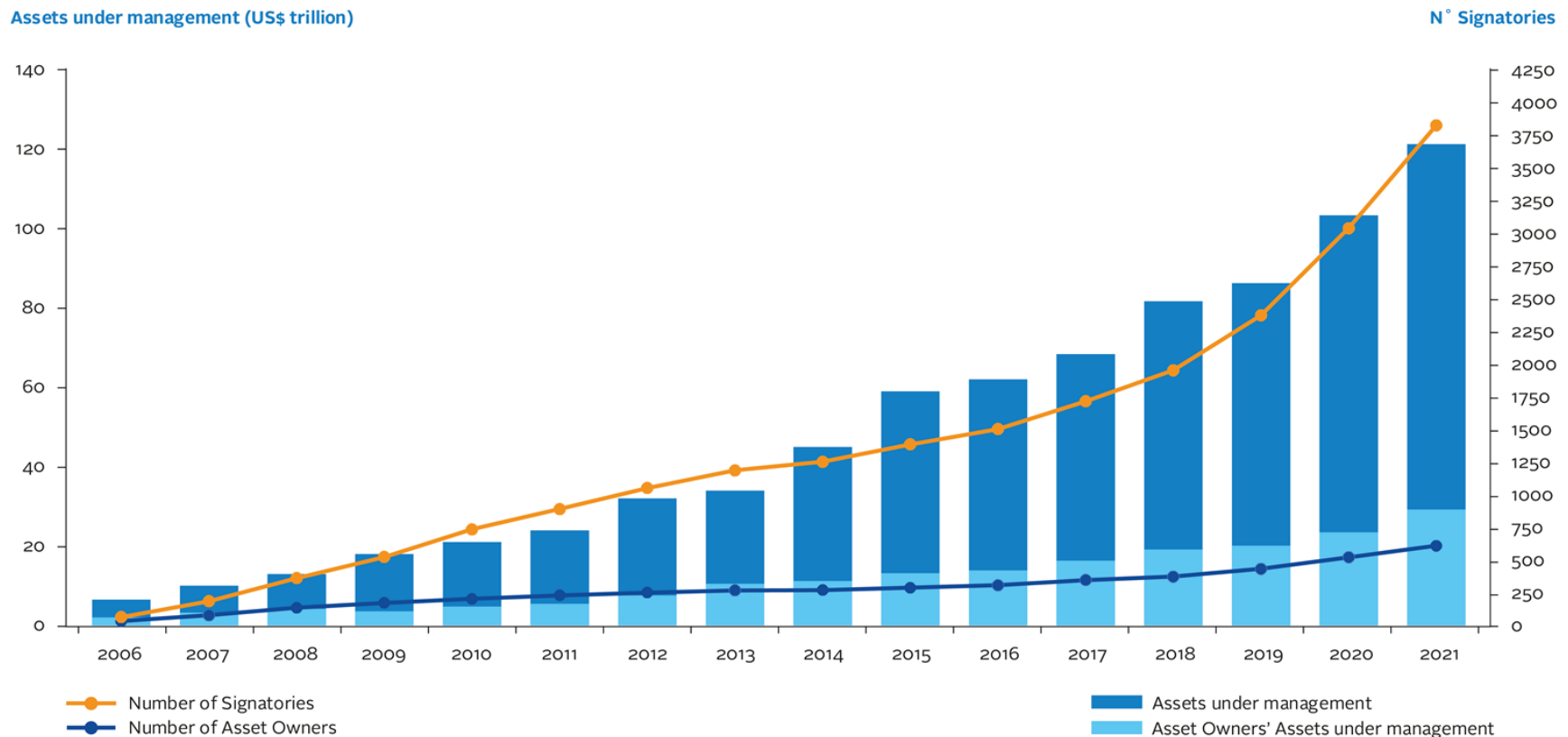
- executive pay
- bribery and corruption
- political lobbying and donations
- board diversity and structure
- tax strategy

Principles for Responsible Investment

<https://www.unpri.org/pri/what-is-responsible-investment>

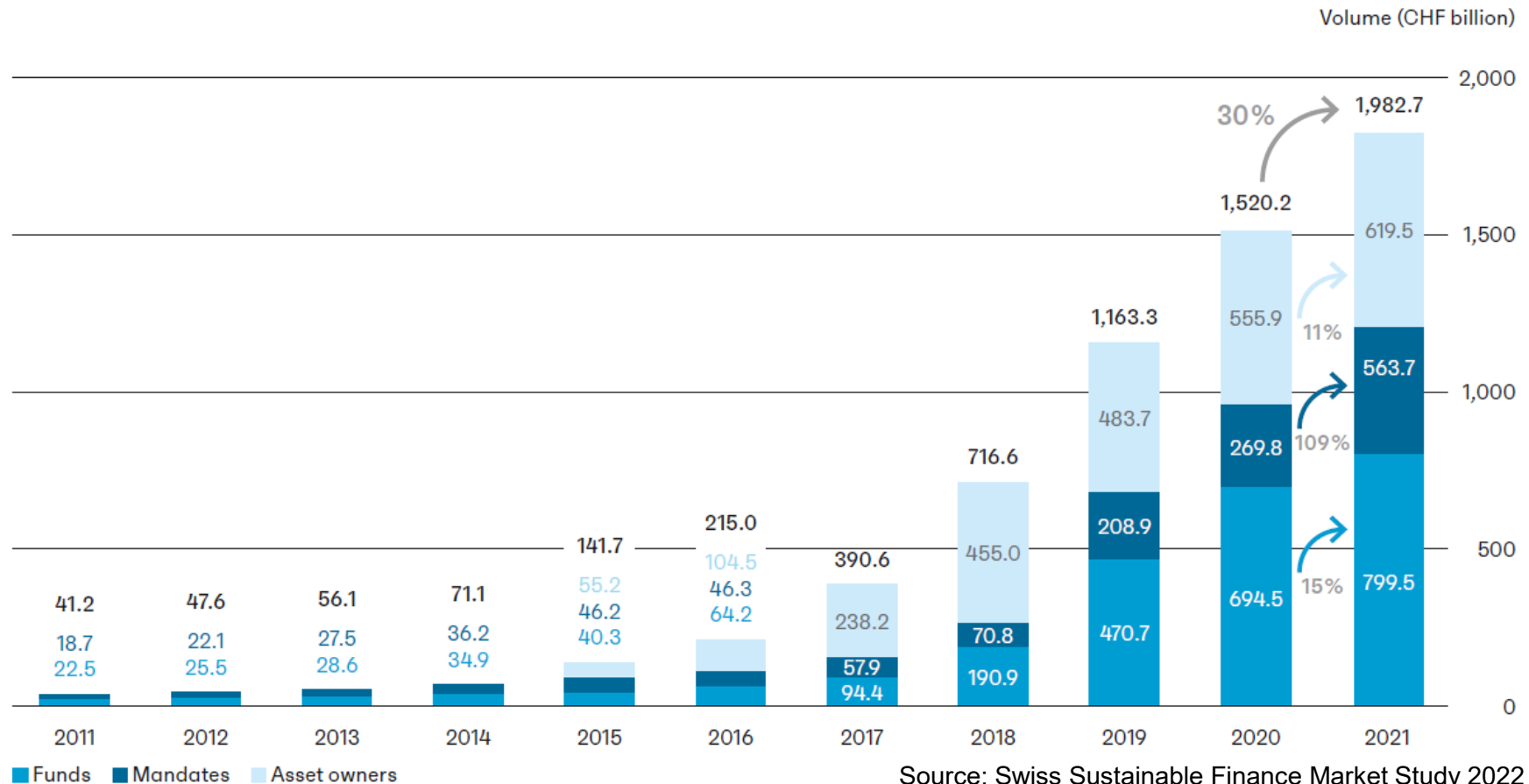
Development of ESG investments

- As of June 2023: 5,372 signatories; USD > 120 trillion AuM

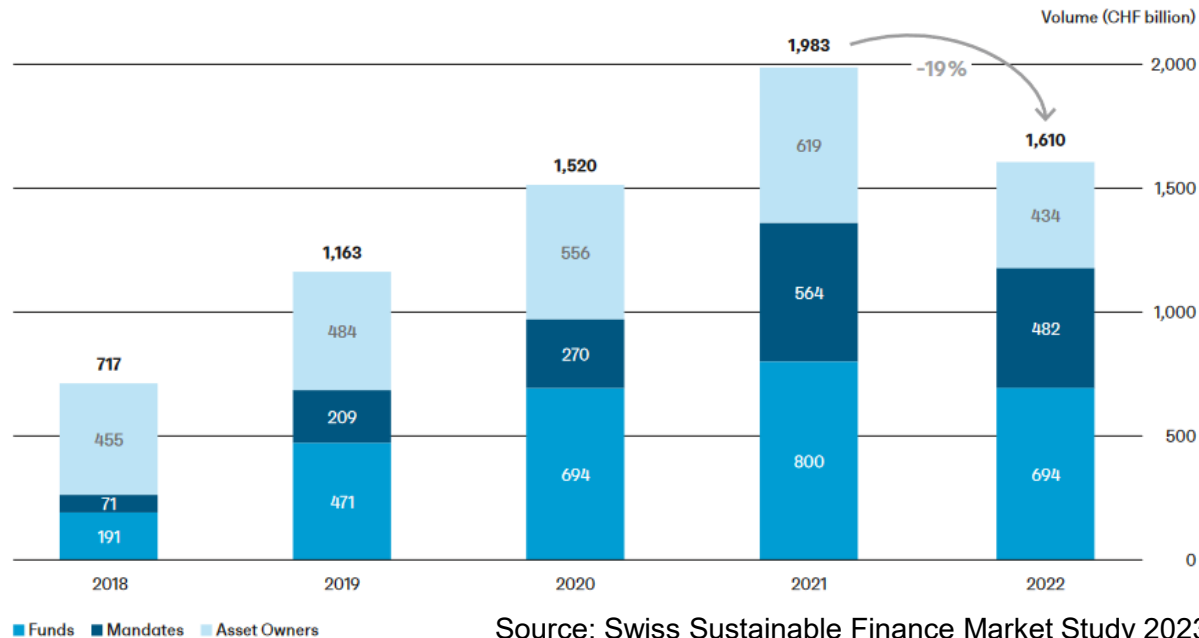


Source: Principles for Responsible Investment, Annual Report 2021

Development of ESG investments in Switzerland



Development of ESG investments in Switzerland



Main drivers of reported decline:

- The main driver (18%) is negative market performance in 2022.
 - Some market players have tightened their definitions of sustainability-related investments managed in Switzerland.
 - A new method for asset owners was implemented which led to lower reported volumes for stewardship approaches.
- ⇒ Net, there were still inflows.

How is ESG measured?

Negative Approach

Exclude firms engaged in controversial business from the investment process

→ e.g., weapons manufacturers, nuclear powerplants, etc.

Positive Approach

Absolute ratings
→ (binary) indicators

Within class ratings
→ (binary) indicators compared with peers (e.g., within industry)

Ratings

- Aggregation of the individual measures associated with the positive approach into a single rating metric
- Usually a number of sub-categories are defined (e.g., human rights, employee relations, diversity, etc.) and these sub-category-ratings are then aggregated into a total rating
- Ratings generally comprise strengths AND concerns of firms.

Who provides ESG data?

MSCI ESG Stats

(formerly KLD Stats)

*1988; USA; discontinued
(acq. in 2009 by RiskMetrics, in
2010 by MSCI)

Inrate

*1990 (Centre Info); *1995 (INrate),
Switzerland

MSCI IVA

(formerly Innovest)

*1992; USA
(acquired in 2009 by RiskMetrics
and in 2010 by MSCI)

Sustainalytics

(since 2019 Morningstar)

*1992; Netherlands

RepRisk

*1998, Switzerland

S&P Global

(formerly RobecoSAM)

*1999, Switzerland

FTSE4Good

*2001; UK

Refinitiv

(formerly Thomson Reuters &
Asset4), USA-UK

*2003; Switzerland

Moody's

(formerly Vigeo Eiris)

*2002; France
(acq. by Moody's in 2019)

ISS ESG Governance

QualityScore

*2013; USA

Truevalue Labs

*2013; USA

(since 2020 owned by FactSet)

Bloomberg ESG

*USA



Problem: Rating divergence

Correlations between ESG ratings at the aggregate rating level (ESG) and at the level of the environmental dimension (E), the social dimension (S), and the governance dimension (G) using the common sample. The results are similar using pairwise common samples based on the full sample. SA, SP, MO, RE, KL, and MS are short for Sustainalytics, S&P Global, Moody's ESG, Refinitiv, KLD, and MSCI, respectively.

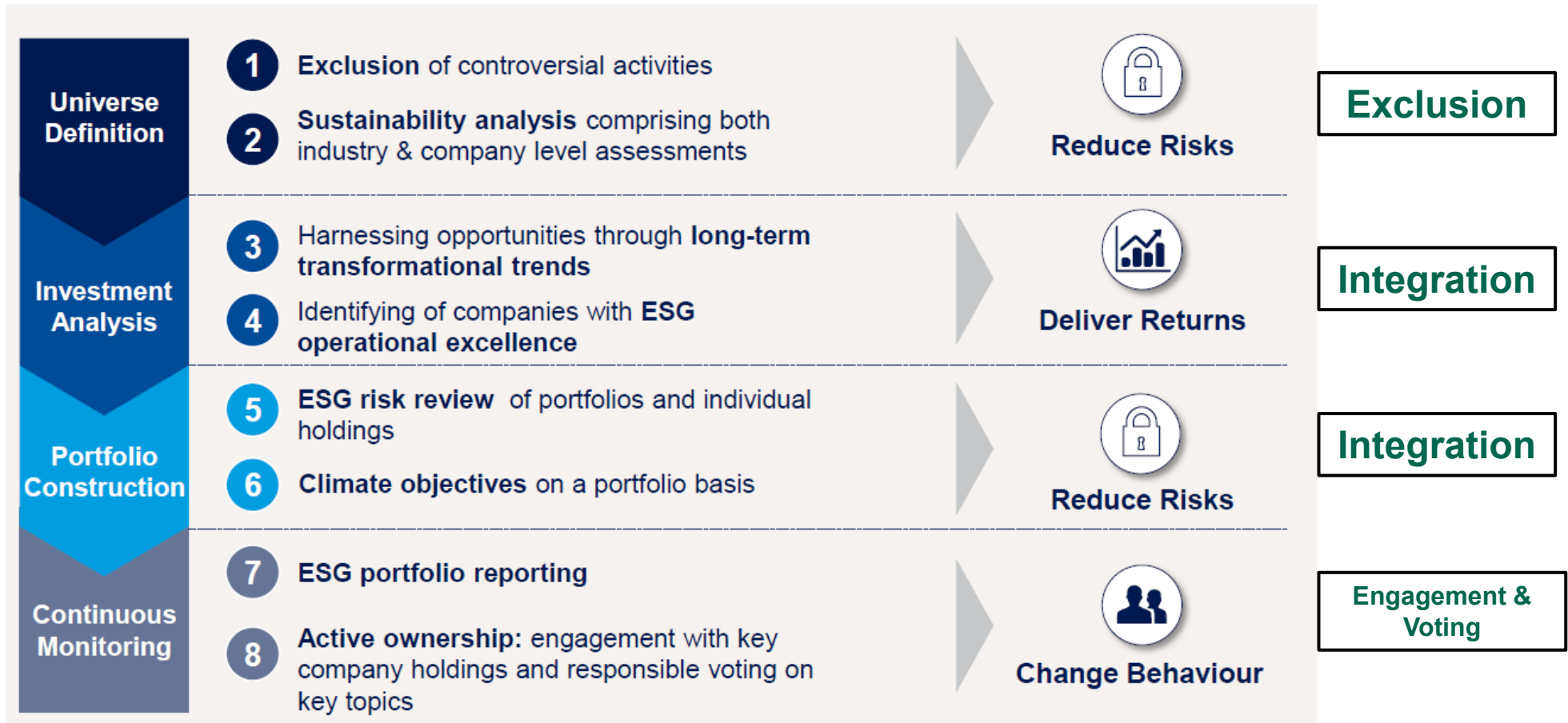
	KL SA	KL MO	KL SP	KL RE	KL MS	SA MO	SA SP	SA RE	SA MS	MO SP	MO RE	MO MS	SP RE	SP MS	RE MS	Average
ESG	0.53	0.49	0.44	0.42	0.53	0.71	0.67	0.67	0.46	0.7	0.69	0.42	0.62	0.38	0.38	0.54
E	0.59	0.55	0.54	0.54	0.37	0.68	0.66	0.64	0.37	0.73	0.66	0.35	0.7	0.29	0.23	0.53
S	0.31	0.33	0.21	0.22	0.41	0.58	0.55	0.55	0.27	0.68	0.66	0.28	0.65	0.26	0.27	0.42
G	0.02	0.01	-0.01	-0.05	0.16	0.54	0.51	0.49	0.16	0.76	0.76	0.14	0.79	0.11	0.07	0.30

Definitions of sustainable investment approaches

Best-in-Class	Approach in which a company's or issuer's ESG performance is compared with that of its peers based on a sustainability rating. All companies or issuers with a rating above a defined threshold are considered as investable.
ESG Engagement	Activity performed by shareholders with the goal of convincing management to take account of ESG criteria so as to improve ESG performance and reduce risks.
ESG Integration	The explicit inclusion by investors of ESG risks and opportunities into traditional financial analysis and investment decisions based on a systematic process and appropriate research sources.
ESG Voting	This refers to investors addressing concerns of ESG issues by actively exercising their voting rights based on ESG principles or an ESG policy.
Exclusions	An approach excluding companies, countries or other issuers based on activities considered not investable. Exclusion criteria (based on norms and values) can refer to product categories (e.g. weapons, tobacco), activities (e.g. animal testing), or business practices (e.g. severe violation of human rights, corruption).
Impact Investing	Investments intended to generate a measurable, beneficial social and environmental impact alongside a financial return. Impact investments can be made in both emerging and developed markets and target a range of returns from below-market to above-market rates, depending upon the circumstances.
Norms-Based Screening	Screening of investments against minimum standards of business practice based on national or international standards and norms.
Sustainable Thematic Investments	Investment in businesses contributing to sustainable solutions, both in environmental or social topics.

Source: Swiss Sustainable Finance Market Study 2021

How can SI strategies be implemented? – An example



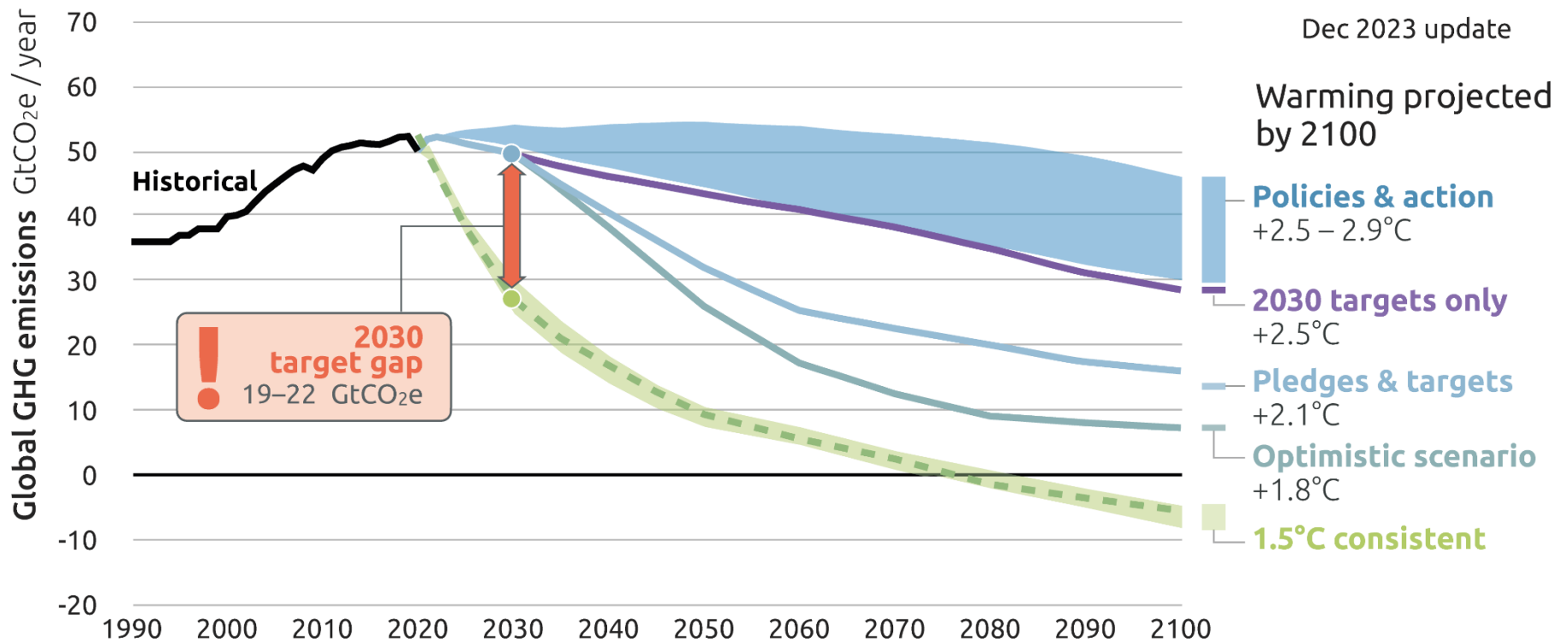
Why is ESG important in corporate finance?

Some empirical evidence

- Demand for ESG investments
- ESG and trust in corporations
- ESG and institutional investors
- Environmental externalities and cost of capital

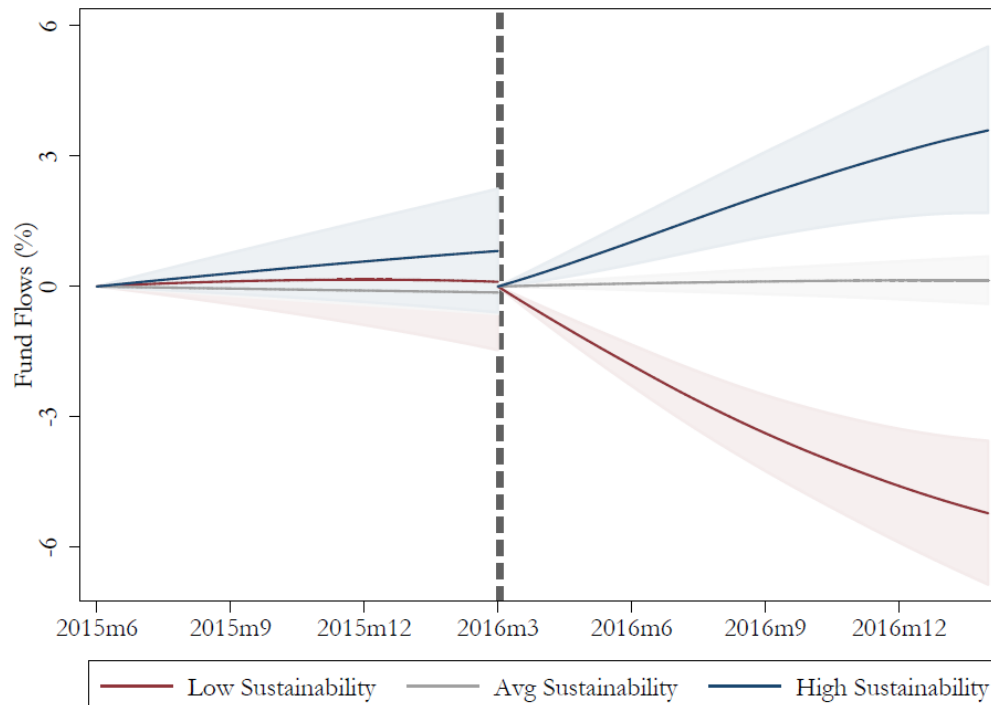
Climate change as main driver of demand for ESG investments

- Warming scenarios by Climate Action Tracker:



Demand for ESG investments

Fund Flows after Sustainability Rating



Source: Hartzmark and Sussman (2019)

- Hartzmark & Sussman (2019) show that investors prefer funds with high sustainability ratings.
- In total, investors withdrew more than \$12 billion from low sustainability funds and invested more than \$24 billion into highly rated funds.
- The authors look at the point in time when Morningstar first published its sustainability ratings.
- However, there is no evidence that high sustainability funds outperform low sustainability funds.

ESG and trust in corporations

- Lins, Servaes, and Tamayo (2017) investigate the impact of corporate social responsibility (CSR) intensity on firm performance during the 2008-2009 financial crisis.
- They focus on the channel that (investment in) social capital can maintain and increase trust in investment firms and hence impact firm performance.
- They explore the financial crisis as exogenous variation in the trust in firms.
- Firms with high CSR (ESG) ratings experienced significantly higher (between four and seven percentage points) stock returns during the financial crisis than those with low CSR ratings. Hence, investors assign a premium to high-CSR firms.
- High-CSR firms had higher profitability and gross margins, and experienced higher sales growth. They were also able to raise more debt during the crisis. These results are consistent with stakeholders and investors considering high-CSR firms to be more trustworthy during the crisis (a time of generally low trust).

ESG and institutional investors

- Dyck, Lins, Roth, and Wagner (2019) assess whether shareholders drive the environmental and social (E&S) performance of firms worldwide.
- Institutional ownership is positively associated with E&S performance, i.e., ESG ratings.
- The effect is even larger after shocks that reveal financial benefits to E&S improvements (e.g., BP Deepwater Horizon oil spill). Firms with higher institutional ownership at the time of the shock are more reactive in improving environmental performance in the following years than firms with low institutional ownership.
- Institutional investors convey their preferences for improved E&S by privately engaging with the firms they already own.
- Foreign institutional investors impact firms' E&S performance only when these investors are from countries with higher demand for E&S performance.

Environmental externalities and cost of capital

- Chava (2014) analyzes the impact of a firm's environmental profile on its cost of (equity and debt) capital.
- Investors demand significantly higher expected returns on stocks excluded by environmental screens compared to firms without such environmental concerns.
- Lenders also charge a significantly higher interest rate on the bank loans issued to firms with environmental concerns.
- Firms with environmental concerns have lower institutional ownership, and fewer banks participate in their loan syndicate than firms without environmental concerns.
- These results suggest that exclusionary socially responsible investing and environmentally sensitive lending can have a material impact on the cost of (equity and debt) capital of affected firms.

Research Paper:

Financing constraints and workplace safety

1. What is the research question of the paper?
2. How is the paper motivated?
3. How is the paper related to our course?
4. What is the relationship between investment and financing constraints?
5. What is the result of the main regression framework?
6. What are potential difficulties in the main regression framework? What are potential solutions to address these problems?
7. What are the authors' strategies in the paper to address these problems?
8. What are the results of these strategies?
9. What do we learn?

Financing constraints and workplace safety

Research question

What is the research question?

Financing constraints and workplace safety

Research question

What is the research question?

- “This paper studies how financing constraints impact workplace safety, which has implications for firm value and employee welfare.” (p. 2017)
 - Do financing constraints of firms increase the likelihood of workplace injuries?
 - What are the costs of workplace injuries to firms?

Financing constraints and workplace safety

Motivation of the paper

How is this paper motivated? Why is it relevant?

Financing constraints and workplace safety

Motivation of the paper

How is this paper motivated? Why is it relevant?

- Significant number of workplace injuries (3.5 million per year) and high associated costs of about \$250 billion → relevant to the **society**
- Firms invest resources into improvements of workplace safety. The investment decision can be affected by financing constraints, because such investments must be financed out of internal cash flows or external capital, and the benefit of such investments is not immediately observable → important for **firms**
- The relationship between financing constraints and investments into workplace safety is largely unexplored → important for **academia**

Financing constraints and workplace safety

Relation to the course

How is the paper related to our course?

Financing constraints and workplace safety

Relation to the course

How is the paper related to our course?

- The **capital structure** and external financing sources can affect the decision to invest.
- **Asymmetric information problem** of investments in workplace safety: The **pecking order theory** suggests difficulties to raise money externally.
- One can imagine that **debt overhang** might play a role for highly leveraged firms. Investments into workplace safety can be profitable, although high levels of leverage restrict firms' financing abilities.
- High injury rates could increase the **cost of capital** as investors require a higher risk premium.
- This paper acts as a good example of how to conduct empirical research.

Financing constraints and workplace safety

Financing constraints

What are financing constraints?

Financing constraints and workplace safety

Financing constraints

What are financing constraints?

- Financing constraints can be characterized in terms of the **curvature** of the supply-of-capital curve: The more inelastic the supply of capital, the more costly the firm finds it to raise an additional unit of external capital. In the limit, the curve becomes vertical, and the firm is shut out of the capital markets.
- Another definition characterizes constraints in terms of the **wedge** between a firm's opportunity cost of internal capital and its cost of external capital. The larger the wedge, the more constrained a firm is.

Financing constraints and workplace safety

Economic intuition

What is the relationship between investment and financing?

Financing constraints and workplace safety

Economic intuition

What is the relationship between investment and financing?

- Cash, i.e., internal financing, can be invested in workplace safety.
- Debt reduces cash flows through interest payments and lowers the ability to raise additional debt.
- Investments, in general, tend to increase with available cash and to decrease in leverage.
- Investments in safety must be financed using either internal cash flows or externally raised capital.
- Investments in safety may be especially vulnerable to be cut in the face of financing constraints, as payoffs accrue slowly over time and are difficult to evaluate ex-ante.

Financing constraints and workplace safety

Investments

What are typical investments to improve workplace safety?

Financing constraints and workplace safety

Investments

What are typical investments to improve workplace safety?

- Investments to improve safety can save employees from injuries.
- Investments into physical equipment
 - Maintaining existing equipment
 - Replacing old and worn out parts and machines
 - Buying equipment with better safety features
 - Automating dangerous tasks
- Investments into less tangible activities
 - Organization of workflow
 - Training and supervision

Financing constraints and workplace safety

Costs & benefits

What are costs and benefits for firms to invest in workplace safety?

Financing constraints and workplace safety

Costs & benefits

What are costs and benefits for firms to invest in workplace safety?

- No systematic estimates of such spending.
- Anecdotal evidence suggests that the amount could be substantial (e.g., 7% of total income). Safety experts agree with such estimates.
- A large number of injuries decreases a firm's productivity due to higher downtimes, paying overtime of other employees, and poor employee morale.
- Higher injury rates can negatively impact firms' reputation, and potential employees ask for higher compensations, i.e., wage differential.

Financing constraints and workplace safety

Dependent variables

What is the main dependent variable?

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Dependent variables

What is the main dependent variable?

- The main dependent variable is Injuries/Hour x 1,000
- This variable measures the overall exposure to injury risk and is ultimately a function of the number of hours that employees spend at work.

Financing constraints and workplace safety

Dependent variables

Are injury rates simply explained by (dangerous) industries?

Financing constraints and workplace safety

Dependent variables

Are injury rates simply explained by (dangerous) industries?

- Please interpret the “Between and Within industry variation in Table IV:
 - Within industry std. dev is three times larger than the Between industry std. dev.
 - There is large variation of injuries within industry and less across industries.
 - This means that there are other important factors explaining injury rates than just specific industries.

Financing constraints and workplace safety

Main empirical results

Please explain the explanatory variables in the main regression specification?

$$\frac{Injuries}{Hour_{ij}} = \alpha_i + \delta_{jt} + \gamma_{st} + \beta X_{it} + \epsilon_{it}$$

Financing constraints and workplace safety

Main empirical results

Please explain the explanatory variables in the main regression specification?

$$\frac{\text{Injuries}}{\text{Hour}_{ij}} = \alpha_i + \delta_{jt} + \gamma_{st} + \beta X_{it} + \epsilon_{it}$$

- βX_{it} : consists of establishment-year and firm-year variables, including those relating to financing
- α_i : Establishment fixed effects account for any time-invariant omitted establishment characteristics
- δ_{jt} : industry-year fixed effects capture the evolution of production technology or industry growth as well as industry-specific shocks
- γ_{st} : state-year fixed effects capture the local labor environment or safety laws

Financing constraints and workplace safety

Main empirical results

Model Dependant variable Sample	OLS Inj/Hour All (1)	OLS Inj/Hour All (2)	OLS Inj/Hour All (3)
Debt/Assets	0.0047** (0.0024)	0.0075** (0.0036)	0.0063** (0.0029)
Cash/Assets	-0.0103** (0.0050)	-0.0171** (0.0068)	-0.0016 (0.0031)
CashFlow/Assets	-0.0051* (0.0029)	-0.0033 (0.0038)	0.0023 (0.0033)
Dividends/Assets	-0.0139* (0.0079)	-0.0098 (0.0121)	-0.0218** (0.0088)
Log(Assets)	-0.0020*** (0.0006)	-0.0010** (0.0005)	-0.0040*** (0.0012)
AssetTurnover	0.0033*** (0.0012)	0.0052*** (0.0015)	0.0005 (0.0010)
MarketToBook	-0.0003 (0.0006)	-0.0004 (0.0007)	0.0002 (0.0004)
TangibleAssetRatio	0.0267*** (0.0072)	0.0376*** (0.0090)	0.0106* (0.0062)
Capex/Assets	-0.0397** (0.0160)	-0.0576** (0.0234)	0.0052 (0.0080)
Log(Employees)	0.0007 (0.0004)	0.0053*** (0.0017)	-0.0006 (0.0009)
Hours/Employee	-0.0000*** (0.0000)	-0.0000 (0.0000)	-0.0000*** (0.0000)
Establishment fixed effects	No	Yes	Yes
Year × Industry dummies	No	No	Yes
Year × State dummies	No	No	Yes
Observations	43,371	43,371	25,053
Adjusted R^2	0.0273	0.0490	0.0618

1. What do the coefficients on the first three variables tell us?
2. What is the economic effect of the coefficient on Debt/Assets?
3. Why are there fewer observations in specification (3) than in (1) and (2)?

Financing constraints and workplace safety

Main empirical results

What do the coefficients on the first three variables tell us?

Financing constraints and workplace safety

Main empirical results

What do the coefficients on the first three variables tell us?

- Columns (1) to (3) show that injury rates are positively related to *Debt/Assets*: In all three columns, the coefficients are significant at the 5% level.
- Column (1) shows that injury rates are negatively related to *Cash/Assets* and *CashFlow/Assets*: However, these relationships lose their significance the more fixed effects are included.
- These results are all consistent with the hypothesis that a better ability to finance investments leads to improved workplace safety and, hence, a lower risk of injury.

Financing constraints and workplace safety

Main empirical results

What is the economic effect of the coefficient on Debt/Assets?

Financing constraints and workplace safety

Main empirical results

What is the economic effect of the coefficient on Debt/Assets?

- The 0.0063 coefficient on *Debt/Assets* in column (3) implies that a one-standard-deviation increase in leverage (0.219) is associated with a 0.0014 unit increase in *Injuries/Hour*.

$$\beta_{Debt/Assets} \cdot \sigma_{Debt/Assets} = \Delta Injuries/Hour$$

$$0.0063 \cdot 0.219 = 0.0014$$

- This represents a modest, though nontrivial, increase of 5.6% relative to the sample mean of *Injuries/Hour* of 0.0247.

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Main empirical results

Why are there fewer observations in specification (3)?

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Main empirical results

Why are there fewer observations in specification (3)?

- Establishment fixed effects capture unobserved, fixed heterogeneity across establishments. In order to include establishment fixed effects, an establishment has to appear in the sample at least twice.
- Otherwise, it is impossible to estimate any establishment effect.
- This reduces the number of observations from 43,371 to 25,053 establishment-years.
- Note: The number of observations in column (2) must also be 25,053, and cannot be 43,371.

Financing constraints and workplace safety

Endogeneity problems

Do you see any endogeneity problems associated with the previous regression framework?

Financing constraints and workplace safety

Endogeneity problems

- Endogeneity can be broadly defined as a situation in which the explanatory variables and the error term in a regression are correlated.
- Sources of endogeneity (e.g., see Roberts and Whited, 2013):
 - Omitted variables – *managerial ability, information asymmetry*
 - Simultaneity – *y and x are jointly determined*
 - Measurement error
- Endogeneity leads to biased and inconsistent parameter estimates. Hence, results may not be meaningful.
- In many cases, endogeneity can be severe enough to reverse even qualitative inference.

Financing constraints and workplace safety

Endogeneity problems

Do you see any endogeneity problems associated with the previous regression framework?

- In general: Many of the explanatory variables are **chosen by firms**. This makes them endogenous by itself as firms jointly decide on many things.
- Other explanations for a potential **negative** relationship
 - Heavy manufacturing equipment makes good collateral to raise debt but simultaneously increases the risk of injuries at work.
 - Poor operational management can increase injury risk while also depleting financial resources.
 - Growing firms may employ less experienced employees with higher injury risk, and growth tends to consume financial resources.
 - Costs associated with actual injuries deplete financial resources.
- Other explanations for a potential **positive** relationship
 - Limited financial resources force firms to operate their business efficiently.
 - Investments into workplace safety when injury rates are high.

Financing constraints and workplace safety

Endogeneity problems

How do the authors address these endogeneity problems?

Financing constraints and workplace safety

Endogeneity problems

How do the authors address these endogeneity problems?

➤ Lead-Lag Analysis

- Column (2) of Table V shows that only the Debt/Asset ratios in the previous two years have a positive effect on injury rates.
- Contemporaneous and future debt/asset ratios have no impact and their magnitudes are close to zero.
- This analysis mostly mitigates concerns of “simultaneity”.

➤ Quasi-natural experiments

- American Jobs Creation Act (AJCA)
- Maturity structure of firms' debt at the onset of the financial crisis in late 2007
- Substantial fluctuations in oil prices

Financing constraints and workplace safety

Quasi-natural experiments

What is the basic idea of a quasi-natural experiment?

Financing constraints and workplace safety

Quasi-natural experiments

What is the basic idea of a quasi-natural experiment?

- Quasi-natural experiments share similarities with a traditional experimental design, or randomized controlled trials, but they specifically lack the element of random assignment to treatment or control.
- Hence, quasi-natural experiments usually use events that are not under the control of the researcher, but exogenous to the subjects (in this case firms). An example could be a new regulation that affects some firms, but not others.
- A major concern to internal validity in a quasi-natural experiment is that treatment and control groups significantly differ already before the treatment.

Financing constraints and workplace safety

Quasi-natural experiments

How would a quasi-natural experiment estimated with a difference-in-differences estimator look like vs. the OLS spec.?

Financing constraints and workplace safety

Quasi-natural experiments

How would a quasi-natural experiment estimated with a difference-in-differences estimator look like vs. the OLS spec.?

Main OLS regression specifications:

$$\frac{\text{Injuries}}{\text{Hour}_{ij}} = \beta \text{Financial Constraints}_{it} + \phi X_{it} \alpha_i + \delta_{jt} + \gamma_{st} + \epsilon_{it}$$

We need exogenous variation in the measures of financial constraints:

$$\frac{\text{Injuries}}{\text{Hour}_{it}} = \beta \text{Exposure}_t \times \text{Treatment}_i + \phi X_{it} + \alpha_i + \delta_{jt} + \gamma_{st} + \epsilon_{it}$$

- Exposure equals 1 for treated establishments and 0 for control establishments
- Treatment equals 1 for observations after the shock and 0 before

Financing constraints and workplace safety

Quasi-natural experiment: AJCA

How does the American Job Creation Act work as a quasi-natural experiment?

Financing constraints and workplace safety

Quasi-natural experiment: AJCA

How does the American Job Creation Act work as a quasi-natural experiment?

- The American Job Creation Act (AJCA) was established in 2004.
- It allowed firms to repatriate foreign income on a one-time basis at a low tax rate of 5.25% instead of the standard corporate tax rate of 35%.
- Firms with profitable foreign subsidiaries benefited most and experienced a significant increase in the level of cash balances.
- Altogether, firms repatriated about \$312 billion to the US.
- This additional income was **unexpected/exogenous** and **significant** for firms.
- Academic literature also shows that firms invested more in response to AJCA.

Financing constraints and workplace safety

Quasi-natural experiment: Debt maturity structure

How can the maturity structure of firms' debt at the onset of the financial crisis be used as a quasi-natural experiment?

Financing constraints and workplace safety

Quasi-natural experiment: Debt maturity structure

How can the maturity structure of firms' debt at the onset of the financial crisis be used as a quasi-natural experiment?

- The main assumption is that managers were not able to anticipate the financial crisis in 2007.
- Under this assumption, the maturity structure of firms is independent of the crisis.
- Those firms with lots of maturing debt during the financial crisis had trouble to roll over this debt. Hence, they were facing constraints.
- The crisis can thus be used as **exogenous variation** in financing constraints for firms with maturing debt.

Financing constraints and workplace safety

Quasi-natural experiment: Oil price fluctuations

How can changes in the oil price be used as a quasi-natural experiment?

Financing constraints and workplace safety

Quasi-natural experiment: Oil price fluctuations

How can changes in the oil price be used as a quasi-natural experiment?

- Oil producers who are also engaged in non-oil businesses can reallocate their capital internally.
- Higher oil prices act as exogenous positive cash flow shocks to the non-oil business of oil producers.
- During the sample period, the price per barrel increased from around \$25 in 2002 to over \$130 per barrel in 2008, before falling to the \$40s in 2009.

Financing constraints and workplace safety

Quasi-natural experiment: AJCA

How do the quasi-natural experiments work empirically?
Let's look at the example of the American Job Creation Act!

Financing constraints and workplace safety

Quasi-natural experiment: AJCA

How do the quasi-natural experiments work empirically?
Let's look at the example of the American Job Creation Act!

- $\frac{\text{Injuries}}{\text{Hour}_{it}} = \beta \text{Treatment}_t \times \text{Exposure}_i + \phi X_{it} + \alpha_i + \delta_{jt} + \gamma_{st} + \epsilon_{it}$
- *Exposure* is a dummy which is equal to 1 if the parent firm's cumulative reported foreign profits are positive, and 0 otherwise. This dummy captures the difference between firms with foreign establishments and firms without.
- *Treatment* is one post-2004 and zero pre-2004. This dummy captures differences which are related to the development over time.
- *Exposure x Treatment* refers to firms with foreign profits after the AJCA was passed.

Financing constraints and workplace safety

Quasi-natural experiment: AJCA

How do the quasi-natural experiments work empirically?
Let's look at the example of the American Job Creation Act!

- This analysis is also called a differences-in-differences analysis.
- The idea is to compare a treatment group (e.g., firms with foreign subsidiaries) with a control group (e.g., firms not having any foreign subsidiaries).
- This framework compares the differences before and after the treatment (e.g., the American Jobs Creation Act) of one group to the other group to get rid of general time trends and cross-sectional differences across the two groups.

Financing constraints and workplace safety

Quasi-natural experiment

Do the quasi-natural experiments support the OLS results?

Financing constraints and workplace safety

Quasi-natural experiment

Do the quasi-natural experiments support the OLS results?

Table VIII – Workplace Injuries and Cash Flow Shocks

Experiment	AJCA (1)	Fin Crisis (2)	Oil Price (3)	AJCA (4)	Oil Price (5)
Treatment $\times$ Exposure	−0.0006* (0.0003)	0.0007** (0.0003)	−0.0008** (0.0004)	0.0006* (0.0004)	−0.0002 (0.0005)
Debt/Assets $\times$ Treatment				−0.0021 (0.0052)	0.0125 (0.0232)
Debt/Assets $\times$ Exposure				−0.0055 (0.0143)	0.0820 (0.1266)
Debt/Assets $\times$ Treatment $\times$ Exposure				−0.0063** (0.0030)	−0.0027* (0.0016)
Observations	3,796	2,300	1,096	3,796	1,096
Adjusted R^2	0.0823	0.1254	0.3794	0.1155	0.3803

Financing constraints and workplace safety

Quasi-natural experiment

Do the quasi-natural experiments support the OLS results?

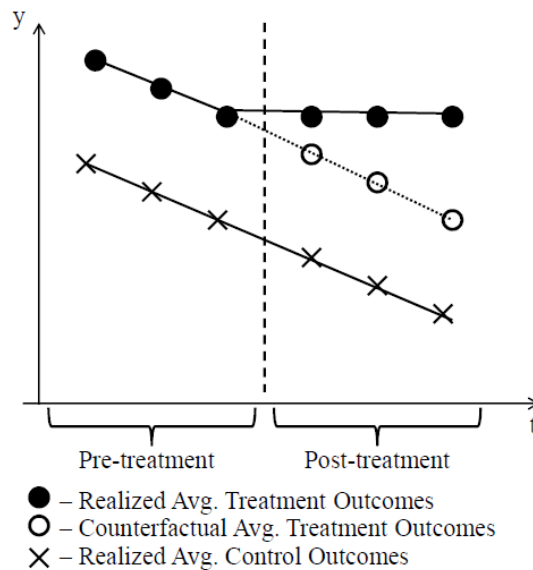
- We find results in Table VIII which are consistent with the main regression results.
- All firms which experienced positive cash flow shocks (*treatment* $\times$ *exposure*) show fewer injury rates.
- Even the triple interaction *Debt/Assets* $\times$ *treatment* $\times$ *exposure* shows consistent negative coefficients suggesting that the effect is larger for firms with more severe financing constraints (i.e., higher debt/assets).

Financing constraints and workplace safety

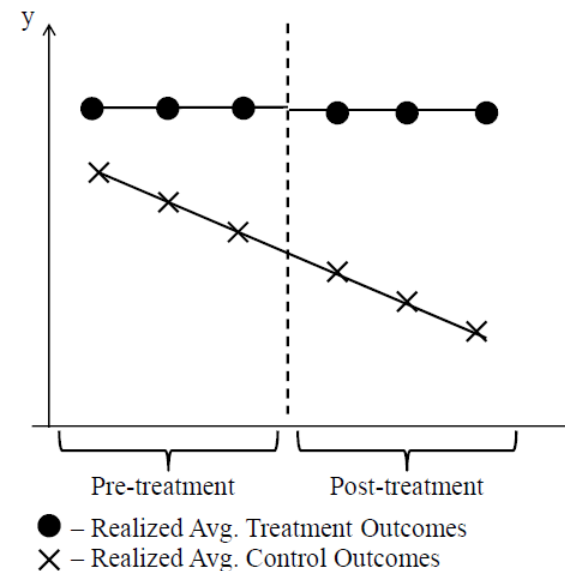
Quasi-natural experiment: Parallel trends

Differences-in-differences analyses heavily rely on the “common trend” assumption, i.e., the dependent variables move in parallel.

Common Trend



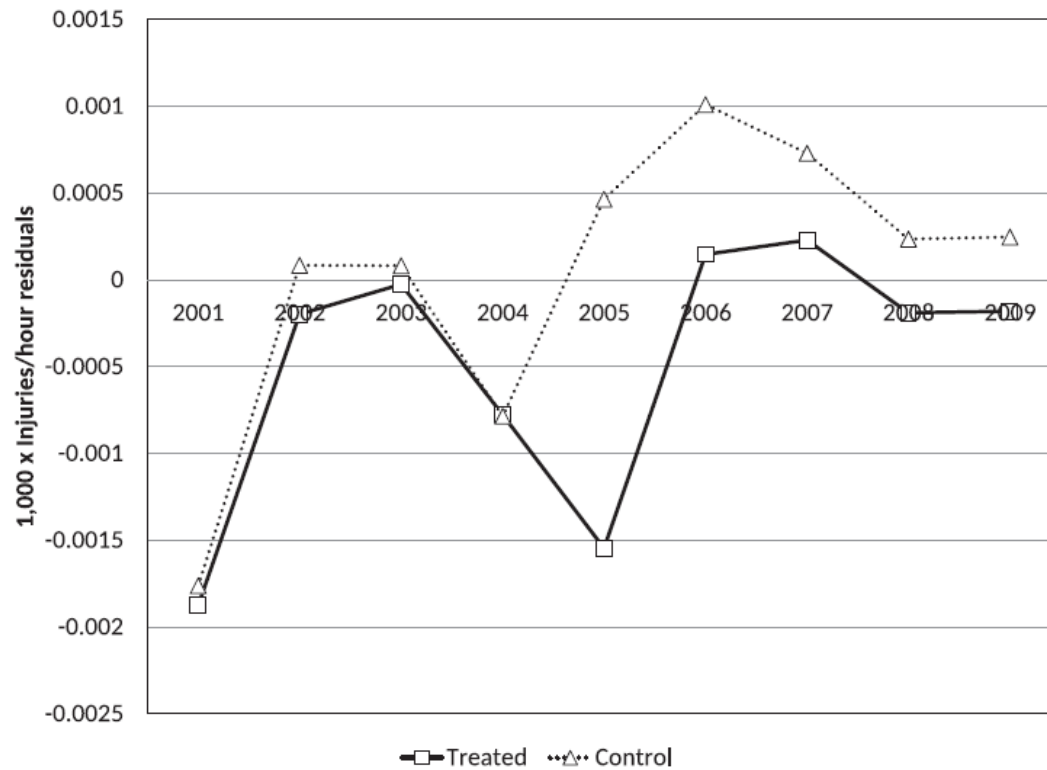
No Common Trend



Financing constraints and workplace safety

Quasi-natural experiment: Parallel trends

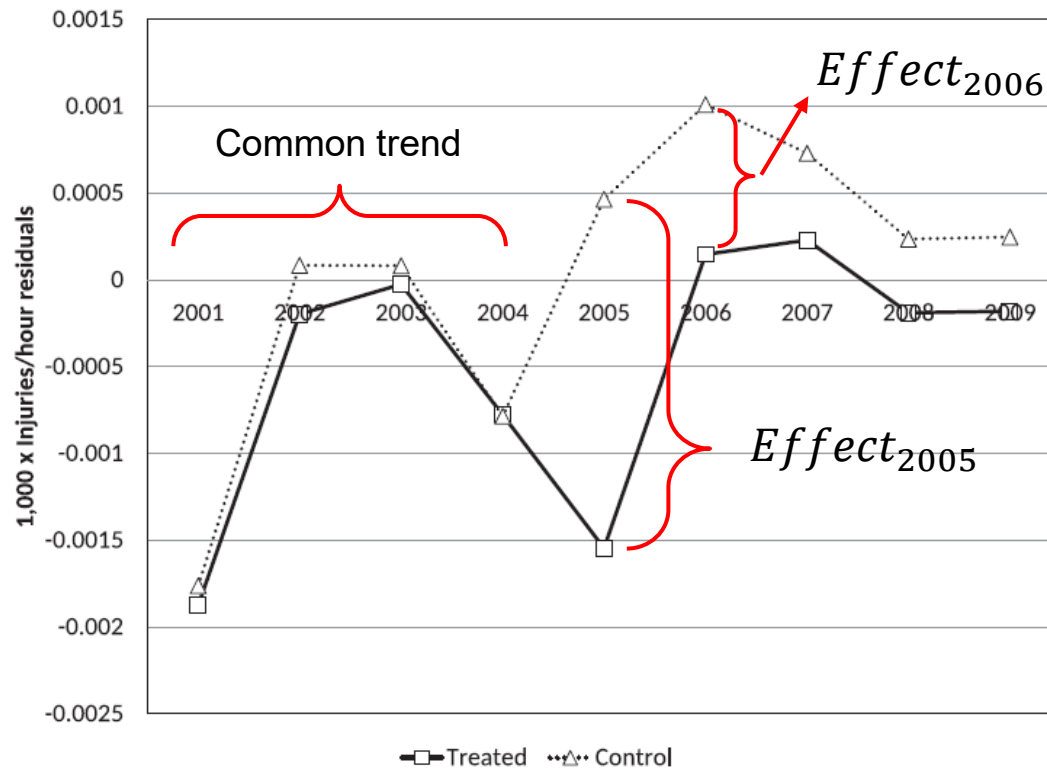
Does the graphical inspection of AJCA support parallel trends?



Financing constraints and workplace safety

Quasi-natural experiment: Parallel trends

Does the graphical inspection of AJCA support parallel trends?

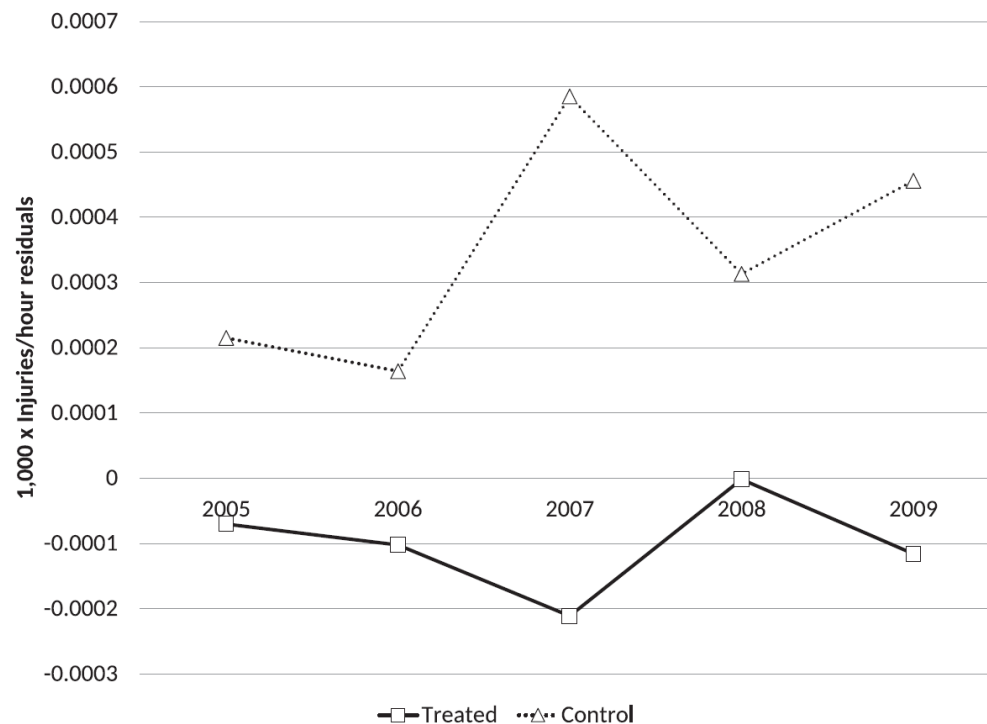


- The graphical inspection of AJCA shows a relatively stable common trend in the years before the introduction of the law.
- There is a significant treatment effect in the first year, i.e., 2005.
- The difference decreases in the following years but persists.

Financing constraints and workplace safety

Quasi-natural experiment: Parallel trends

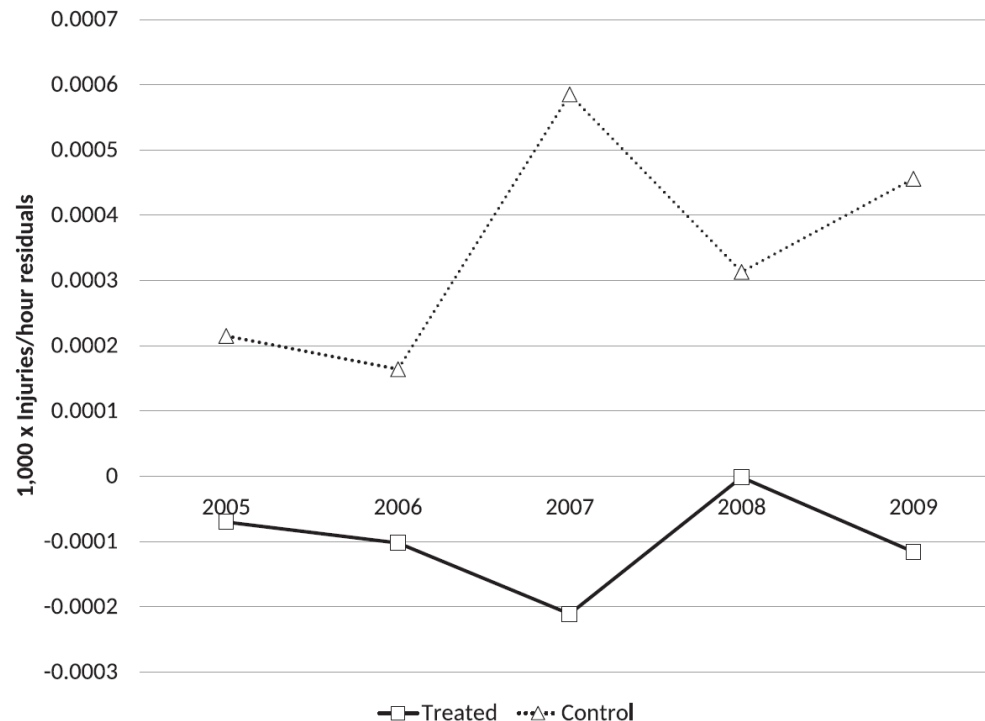
Does the graphical inspection of the maturity structure of debt at the onset of the financial crisis support results?



Financing constraints and workplace safety

Quasi-natural experiment: Parallel trends

Does the graphical inspection of the maturity structure of debt at the onset of the financial crisis support results?



- The graphical inspection of the maturity structure of debt shows a common trend in 2005 and 2006.
- However, the effect of the financial crisis seems to be stronger for the control group.
- The common trend assumption is hence questionable.

Financing constraints and workplace safety

Firm value and injury rates

Do firms have to bear any costs from higher injury rates?

Financing constraints and workplace safety

Firm value and injury rates

Do firms have to bear any costs from higher injury rates?

Table X – Firm Value and Workplace Injuries

	(1)	(2)	(3)	(4)
LaggedInjuries/Hour	-3.190*** (1.164)	-2.715** (1.129)	-3.524** (1.452)	-3.079** (1.369)
LeadInjuries/Hour			-0.414 (0.95)	-0.036 (0.871)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,469	4,469	2,843	2,843
Adjusted R ²	0.1	0.1571	0.1308	0.2021

Financing constraints and workplace safety

Firm value and injury rates

Do firms have to bear any costs from higher injury rates?

- Table X shows results of the regression of Tobin's Q on LaggedInjuries/Hour, LeadInjuries/Hour and additional control variables.
- The coefficients of LaggedInjuries/Hour are consistently negative and indicate that firm value decreases in pre-year injury rates.
- In contrast, the coefficients of LeadInjuries/Hour are insignificant, mitigating the concern of reverse causality.
- The economic magnitude of the impact of lagged injuries is large. A one-standard-deviation increase in injury rate (0.0225) decreases firm value by 6.1% ($0.0225 \times (-2.715) = 0.061$).

Recap

- This paper provides new evidence that injury rates increase with leverage and decrease with operating cash flow.
- Financing can affect the well-being of employees through limited investments in workplace safety.
- The results also have implications for optimal financial policy, as firm value appears to decline with injury rates.

Literature

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Additional literature

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